

Nikita Tsoy — Resume

Sofia – Bulgaria

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Education

INSAIT, Sofia University

PhD in ML

Advisor: N. Konstantinov. Topics: Trustworthy ML

Sofia, Bulgaria

2023–present

New Economic School

MA in Economics, GPA: 5.1/5.3

Thesis: Radio Impact on the US Banking Crisis of the 1930s

Moscow, Russia

2020–2022

Moscow Institute of Physics and Technology

BS in Physics, GPA: 4.9/5

Thesis: Analogue of Chernoff Theorem for Cylindrical Pseudomeasures

Moscow, Russia

2016–2020

Publications

- 1: **Tsoy, N.**, & Konstantinov, N. (2023). Strategic Data Sharing between Competitors. *NeurIPS*.
- 2: **Tsoy, N.**, Mihalkova, A., Todorova, T. N., & Konstantinov, N. (2024). Provable Mutual Benefits from Federated Learning in Privacy-Sensitive Domains. *AISTATS*.
- 3: **Tsoy, N.**, & Konstantinov, N. (2024). Simplicity Bias of Two-Layer Networks beyond Linearly Separable Data. *ICML*.
- 4: **Tsoy, N.**, Kirev, I., Rahimiyazdi, N., & Konstantinov, N. (2025). On the Impact of Performative Risk Minimization for Binary Random Variables. *ICML*.

Working papers

- 1: **Tsoy, N.**, & Konstantinov, N. (2025). Measuring the Layer-Wise Impact of Image Shortcuts on Deep Model Features. *SCSL Workshop*.
- 2: Chakarov, D., **Tsoy, N.**, Minchev, K., & Konstantinov, N. (2024). Incentivizing Truthful Collaboration in Heterogeneous Federated Learning. *OPT Workshop*.
- 3: Campbell, D., Brodeur, A., Dreber, A., Johannesson, M., Kopecky, J., Lusher, L., & **Tsoy, N.** (2024). The Robustness Reproducibility of the American Economic Review. *Preprint*.

Experience

ETH Zurich	Zurich, Switzerland
<i>RA in ML</i>	2022–2023
Investigated the impact of market structure on data sharing incentives	
New Economic School	Moscow, Russia
<i>TA in Mathematics</i>	2021
Conducted seminars on mathematical methods in economics	
Moscow Institute of Physics and Technology	Moscow, Russia
<i>TA in Mathematics</i>	2020
Checked calculus homework	
Moscow Institute of Physics and Technology	Moscow, Russia
<i>RA in Physics</i>	2019–2020
Investigated matrix models in theoretical physics	

Skills

Methods: econometrics, optimization, time series, numerical methods

Finance: corporate finance, asset pricing, mathematical finance, macroeconomic modeling

Machine Learning: game theory, shortcut mitigation, differential privacy

Programming: Python, PyTorch, numpy, pandas, Stata, R, C++

Languages: Russian (native), English (professional working proficiency)